Code Coverage Report

Online Bookstore – SE Mini-Project (UE24CS341A), PES University, Group 2

Part 3 – Testing (2–6 November 2026)

Item	Detail
Project	Online Bookstore (multi-seller book marketplace)
Prepared by	Group 2
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Date	6 November 2026
Repository	SE-OnlineBookstore/SE-Mini-Project
Version	1.0

1. Purpose and scope

This report gives the statement, branch, function and line coverage of the server code produced by the automated tests (Jest) with "npm run test:coverage" on the final main branch on 6 November 2026. Coverage is collected from the services, middleware and utils folders. The client has no automated tests; it is covered by system validation.

2. Summary

Measure	Result
Test suites	33 passed, 33 total
Tests	256 passed, 256 total
Run time	270.94 s (run in band)
Statements	88.81%
Branches	74.56%
Functions	90.67%
Lines	90.1%
NFR-06 gate (services statements at least 70%)	Met: 88.61%

3. Coverage by area

Area	% Stmts	% Branch	% Funcs	% Lines
All files	88.81	74.56	90.67	90.1
middleware	95.23	85.45	100	94.59
services	88.61	73.71	91.19	89.94
utils	82.97	78.94	82.6	86.84

4. Coverage by file

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered lines
middleware	95.23	85.45	100	94.59	
authorizeRoles.js	100	100	100	100	
errorHandler.js	81.81	80	100	80.95	11,15,17,24
rateLimit.js	100	66.66	100	100	23
requestId.js	100	50	100	100	6
validate.js	100	90	100	100	11
verifyJWT.js	100	100	100	100	
services	88.61	73.71	91.19	89.94	
analyticsService.js	100	100	100	100	
auditService.js	100	100	100	100	
authService.js	93.33	82.14	100	96.96	63-64
cartService.js	79.71	66.66	83.33	80.88	77,85,94,98,102-103,120,125,129,146,154-156
catalogueService.js	90.82	75.94	89.47	95.4	72,143,164-165
checkoutService.js	88.97	70.4	100	92.85	50,73,87,98-99,112,175,214,222
fulfilmentService.js	96.29	84.44	100	96.22	35,75
invoiceService.js	95.42	69.23	88.88	96	229,258,263-264,295-296
listingService.js	89.09	79.06	100	89.09	15,38,67,87,113,158
moderationService.js	91.3	75	100	91.3	28,35,42,49
notificationService.js	92.85	80	100	92.85	140-145
orderService.js	68.22	61.15	57.14	69.18	19-25,31,40,47,57,61,65,94,99,103,107,112-113,161-284,361,372,427,477
otpService.js	100	90.9	100	100	15,30
paymentService.js	88.61	69.69	100	89.83	93,98,133,179,194-197,227,237,242,261
reviewService.js	90.9	85.71	100	92.59	45-46
sellerDashboardService.js	100	100	100	100	
tokenService.js	96.96	85.71	100	100	34,56
userService.js	100	95.45	100	100	70
wishlistService.js	86.3	68.18	85.71	87.5	67,74,96,114,133,136,176,186

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utils	82.97	78.94	82.6	86.84	
AppError.js	93.75	70	90.9	93.75	21
asyncHandler.js	100	100	100	100	
crypto.js	100	66.66	100	100	4
isbn.js	100	100	100	100	
money.js	0	100	0	0	2-6

5. Change during Part 3

Measure	Run of 5 Nov (234 tests)	Final run of 6 Nov (256 tests)
Statements	87.18	88.81
Branches	72.38	74.56
Functions	89.11	90.67
Lines	88.58	90.10
Services statements	86.83	88.61

New tests added in this period: auditService (record, failure path, list, paging, filter, append-only API; 50% to 100% statements), userService (profile and address book rules; 65.95% to 100% statements) and order API tests added with the My Orders page (PR #90).

6. Files below 80% and the reason

File	% Stmts	Uncovered lines	Status
cartService.js	79.71	77,85,94,98,102-103,120,125,129,146,154-156	Remaining paths not covered by the current tests; recorded as known gap.
orderService.js	68.22	19-25,31,40,47,57,61,65,94,99,103,107,112-113,161-284,361,372,427,477	Remaining paths not covered by the current tests; recorded as known gap.
money.js	0	2-6	money.js is not used by the application; candidate for removal.

7. Conclusion

Overall statement coverage is 88.81% and every service meets the NFR-06 gate individually except orderService.js (68.22%), whose shortfall is listed above. The services folder as a whole is at 88.61%, above the 70% target.